

Marmara University, 2020

Modeling and Discrete Simulation

Subject 9

Verification and Validation of Simulation Models

Mujdat Soyturk, Ph.D.
Associate Professor

Purpose & Overview



- The goal of the validation process is:
  - To produce a model that represents true behavior closely enough for decision-making purposes
  - To increase the model's credibility to an acceptable level
- Validation is an integral part of model development
  - Verification – building the model correctly (correctly implemented with good input and structure)
  - Validation – building the correct model (an accurate representation of the real system)
- Most methods are informal subjective comparisons while a few are formal statistical procedures

Slides are derived from the course textbook "Banks, Carson, Nelson, Nicol. Discrete-Event System Simulation, 5th Edition, Prentice Hall, 2010" and from textbook's course slides.

Verification vs. Validation

- *Verification*
  - Computational model should be consistent with specification model
  - Did we build the model right?

Verification vs. Validation

- *Validation*
  - Computational model should be consistent with the system being analyzed
  - Did we build the right model?
  - Can an expert distinguish simulation output from system output?

Contents

- Modeling-Building
- Verification
- Validation

Slides are derived from the course textbook "Banks, Carson, Nelson, Nicol. Discrete-Event System Simulation, 5th Edition, Prentice Hall, 2010" and from textbook's course slides.

MODELING-BUILDING

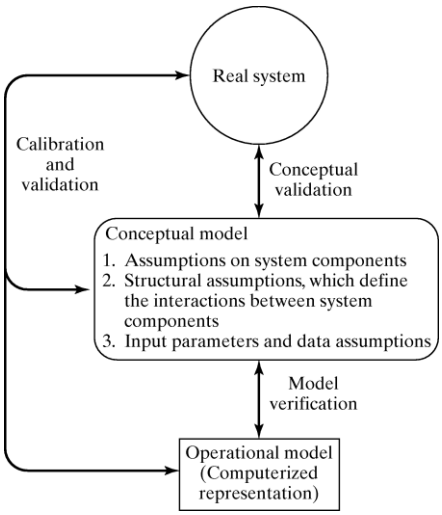
Model Building

Three steps:

- **Observing the real system and the interactions** among its various components and of collecting data on their behavior.
  - This step can be repeated many times during the construction of conceptual model.
- **Construction of conceptual model** – a collection of assumptions about the components and the structure of the system, plus the hypotheses about the values of model input parameters.
- **Implementation of an operational model**, usually by using simulation software, and incorporating the assumptions of the conceptual model into the worldview and concepts of the simulation software.

Model builder may return to each of these steps many times while building, verifying, and validating the model.

Modeling-Building, Verification & Validation



VERIFICATION

Verification

- Purpose: ensure the conceptual model is reflected accurately in the computerized representation.
- In the verification, many common-sense suggestions:
  - Have someone else check the model (preferably an expert).
  - Make a flow diagram that includes each logically possible action a system can take when an event occurs.
  - Closely examine the model output for reasonableness under a variety of input parameter settings. **(Often overlooked!)**
  - Print the input parameters at the end of the simulation, make sure they have not been changed inadvertently.
  - Make the operational model as self-documenting as possible.
  - If there is animation, verify that animation imitates the actual system.
  - Use graphical interfaces if possible.

Examination of Model Output for Reasonableness

- Example: A model of a complex network of queues consisting many service centers.
  - Response time is the primary interest, however, it is important to collect and print out **many statistics in addition to response time**.
    - Two statistics that give a quick indication of model reasonableness are **current contents** and **total counts**, for example:
      - If the current content grows in a more or less linear fashion as the simulation run time increases, it is likely that a queue is unstable
      - If the total count for some subsystem is zero, indicates no items entered that subsystem, a highly suspect occurrence
      - If the total and current count are equal to one, can indicate that an entity has captured a resource but never freed that resource.
    - Output checking and forecasting the reasonable range of the output statistics will aid in error detection.
    - Compute certain long-run measures of performance, e.g. compute the long-run server utilization and compare to simulation results (**analytical vs. simulation**)

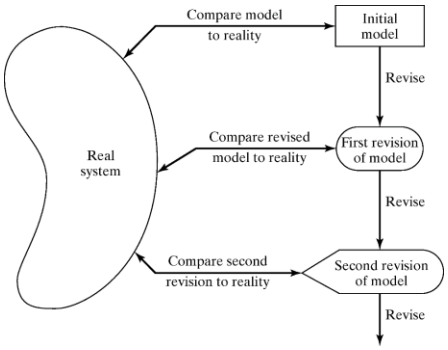
Other Important Tools

- Documentation
  - A means of clarifying the logic of a model and verifying its completeness
- Use of a trace
  - A detailed printout of the state of the simulation model over time.

CALIBRATION & VALIDATION

Calibration and Validation

- Validation: the overall process of comparing the model and its behavior to the real system.
- Calibration: the iterative process of comparing the model to the real system and making adjustments.



Calibration and Validation

- No model is ever a perfect representation of the system
  - The modeler must weigh the possible, but not guaranteed, ***increase in model accuracy*** versus ***the cost of increased validation effort***.
  - Define maximum acceptable discrepancy.
- Three-step approach:
  - Build a model that has high face validity.
  - Validate model assumptions.
  - Compare the model input-output transformations with the real system's data.

High Face Validity

- Ensure a high degree of realism: Potential users should be involved in model construction (from its conceptualization to its implementation).
- Sensitivity analysis can also be used to check a model's face validity.
  - By model user and or model builder: check whether the model behaves in the expected way when one or more input parameters is changed.
    - Example: In most queuing systems, if the arrival rate of customers were to increase, it would be expected that server utilization, queue length and delays would tend to increase.
  - For large-scale simulation models (if too expensive or time consuming to test all possible varying values of input variables), the model builder must choose the most critical input variables for testing.

Validate Model Assumptions

- General classes of model assumptions:
  - **Structural assumptions**: how the system operates.
  - **Data assumptions**: reliability of data and its statistical analysis.
- Bank example: customer queuing and service facility in a bank.
  - Structural assumptions, e.g., customer waiting in one line versus many lines, served FCFS versus priority.
  - Data assumptions, e.g., interarrival time of customers, service times for commercial accounts.
    - Verify data reliability with bank managers.
    - Test correlation and goodness of fit for data (see notes at previous lectures for more details).

Validate Input-Output Transformation

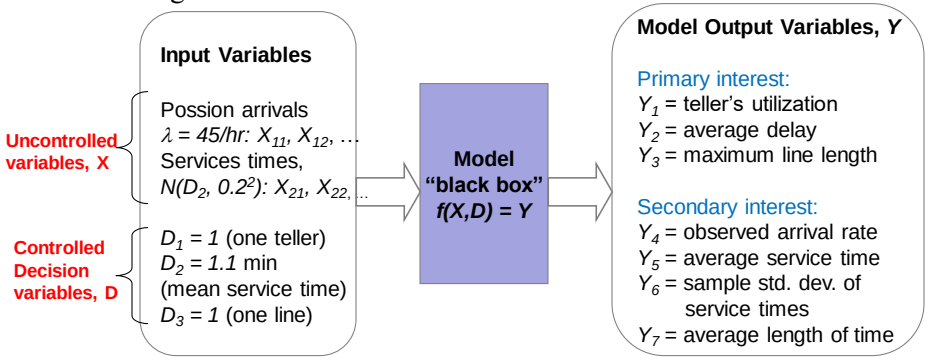
- Goal: Validate the model's ability to predict future behavior
  - The only objective test of the model.
  - The structure of the model should be accurate enough to make good predictions for the range of input data sets of interest.
- One possible approach: use historical data that have been reserved for validation purposes only.
- Criteria: use the main responses of interest.

Bank Example

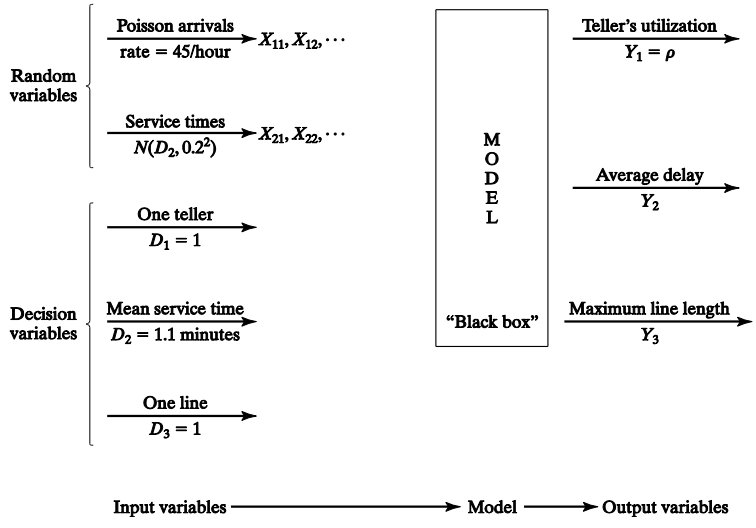
- Example: One drive-in window serviced by one teller, only one or two transactions are allowed.
  - Data collection: 90 customers during 11 am to 1 pm.
    - Observed service times $\{S_i, i = 1,2, \ldots, 90\}$.
    - Observed interarrival times $\{A_i, i = 1,2, \ldots, 90\}$.
  - Data analysis let to the conclusion that: (the input variables)
    - (Interarrival times: exponentially distributed) → Arrivals: Poisson distributed with rate $\lambda = 45$
    - Service times: $N(1.1, 0.2^2)$

The Black Box

- A model was developed in close consultation with bank management and employees
- Model assumptions were validated
- Resulting model is now viewed as a "black box":



The Black Box



Comparison with Real System Data - 1

- Real system data are necessary for validation.
  - System responses should have been collected during the same time period (e.g. in the given example, from *11*am to *1*pm on the same Friday.)
- Compare the average delay from the model Y_2 with the actual delay Z_2:
  - Average delay observed, $Z_2 = 4.3$ minutes, consider this to be the true mean value $\mu_0 = 4.3$.
  - When the model is run with generated random variates X_{1n} and X_{2n}, Y_2 should be close to Z_2.

Comparison with Real System Data - 2

- The generated input values X_{1n} and X_{2n} *cannot be expected to replicate the actual input values A_n and S_n of the real system exactly, but they are expected to replicate the statistical pattern of the actual inputs*.
- Hence, simulation-generated values of Y_2 are expected to be consistent with the observed system variable Z_2.
- Modeler must do tests for this consistency.
  - Modeler makes a small number of statistically independent replications of the model.
  - Then, does the statistical analysis of the values of the output variables.
    - e.g. in the given example, Six statistically independent replications of the model, each of *2*-hour duration, are run.

Hypothesis Testing

- Compare the average delay from the model Y_2 with the actual delay Z_2 (continued):
  - Null hypothesis testing: evaluate whether the simulation and the real system are *the same* (w.r.t. output measures):

$$H_0: E(Y_2) = 4.3 \text{ minutes}$$
$$H_1: E(Y_2) \neq 4.3 \text{ minutes}$$

    - If H_0 is not rejected, then, there is no reason to consider the model invalid
    - If H_0 is rejected, the current version of the model is rejected, and the modeler needs to improve the model

Hypothesis Testing

– Conduct the *t* test:

- Chose level of significance ($\alpha = 0.05$) and sample size ($n = 6$), see result in Table 10.2.

Table 10.2

Replication	Y_4 (Arrivals/Hour)	Y_5 (Minutes)	Y_2 = Average Delay (Minutes)
1	51	1.07	2.79
2	40	1.12	1.12
3	45.5	1.06	2.24
4	50.5	1.10	3.45
5	53	1.09	3.13
6	49	1.07	2.38
Sample mean			2.51
Standard deviation			0.82

- Compute the same mean and sample standard deviation over the n replications:

$$\bar{Y}_2 = \frac{1}{n}\sum_{i=1}^{n} Y_{2i} = 2.51 \text{ minutes} \qquad S = \sqrt{\frac{\sum_{i=1}^{n}(Y_{2i} - \bar{Y}_2)^2}{n-1}} = 0.82 \text{ minutes}$$

Hypothesis Testing

– Conduct the *t* test:

- Chose level of significance ($\alpha = 0.05$) and sample size ($n = 6$), see result in Table 10.2.
- Compute the same mean and sample standard deviation over the n replications:

$$\bar{Y}_2 = \frac{1}{n}\sum_{i=1}^{n} Y_{2i} = 2.51 \text{ minutes} \qquad S = \sqrt{\frac{\sum_{i=1}^{n}(Y_{2i} - \bar{Y}_2)^2}{n-1}} = 0.82 \text{ minutes}$$

- Compute test statistics:

$$|t_0| = \left|\frac{\bar{Y}_2 - \mu_0}{S/\sqrt{n}}\right| = \left|\frac{2.51 - 4.3}{0.82/\sqrt{6}}\right| = 5.34 > t_{critical} = 2.571 \text{ (for a 2-sided test)}$$

- Hence, reject H_0. Conclude that the model is inadequate.
- Check: the assumptions justifying a *t* test, that the observations (Y_{2i}) are normally and independently distributed.

Hypothesis Testing

- Similarly, compare the model output with the observed output for other measures:
  $Y_4 \leftrightarrow Z_4$, $Y_5 \leftrightarrow Z_5$, and $Y_6 \leftrightarrow Z_6$

Hypothesis Testing - Updated

- It is found that an assumption was wrong. After updating the assumptions and justifying the input parameters, new results are obtained.

Table 10.3

Replication	Y_4 (Arrivals/Hour)	Y_5 (Minutes)	Y_2 = Average Delay (Minutes)
1	51	1.07	5.37
2	40	1.11	1.98
3	45.5	1.06	5.29
4	50.5	1.09	3.82
5	53	1.08	6.74
6	49	1.08	5.49
Sample mean			4.78
Standard deviation			1.66

Hypothesis Testing - Updated

- Conduct the *t* test, again:
  - Compute the same mean and sample standard deviation over the n replications:

$$\bar{Y}_2 = \frac{1}{n}\sum_{i=1}^{n} Y_{2i} = 4.78 \text{ minutes}$$

$$S = \sqrt{\frac{\sum_{i=1}^{n}(Y_{2i} - \bar{Y}_2)^2}{n-1}} = 1.66 \text{ minutes}$$

  - Compute test statistics:

$$|t_0| = \left|\frac{\bar{Y}_2 - \mu_0}{S/\sqrt{n}}\right| = \left|\frac{\mathbf{4.78} - 4.3}{\mathbf{1.66}/\sqrt{6}}\right| = \mathbf{0.710} < t_{critical} = 2.571 \text{ (for a 2-sided test)}$$

  - Hence, **do not reject H_0**. Conclude that the model is adequate.
  - ***Should we do more test?***

Type II Error

- For validation, the ***power of the test*** is:
  - **Probability[**detecting an invalid model**]** = $1 - \beta$
  - β = *P(Type II error)* = P(failing to reject $H_0|H_1$ is true)
  - Consider failure to reject H_0 as a strong conclusion, the modeler would want β to be small.
  - Value of β depends on:
    - Sample size, *n*
    - The true difference, δ, between *E(Y)* and μ: $\delta = \frac{|E(Y) - \mu|}{\sigma}$
- In general, the best approach to control β error is:
  - Specify the critical difference, δ.
  - Choose a sample size, *n*, by making use of the operating characteristics curve (OC curve).

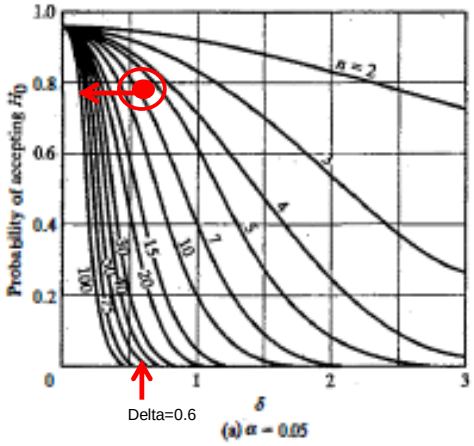
Type II Error

Table A.10 Operating Characteristic Curves for The Two-Sided *t* Test for Different Values of Sample Size *n*

$$\delta = \frac{|E(Y) - \mu|}{\sigma}$$

$$\delta = \frac{|E(Y) - 4.3|}{1.66}$$

$$\delta = \frac{1}{1.66} = 0.6$$



Type I and II Error

- Type I error (α):
  - Error of rejecting a valid model.
  - Controlled by specifying a small level of significance α.
- Type II error (β):
  - Error of accepting a model as valid when it is invalid.
  - Controlled by specifying critical difference and find the n.
- For a fixed sample size n, increasing α will decrease β.

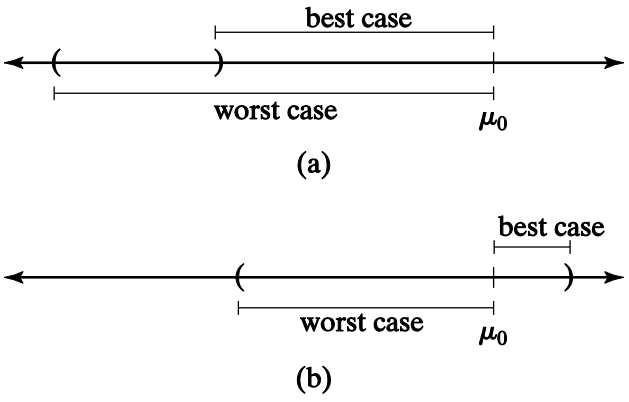
Confidence Interval Testing

- Confidence interval testing: evaluate whether the simulation and the real system are close enough.
- If Y is the simulation output, and $\mu = E(Y)$, the confidence interval (C.I.) for μ is:

$$\bar{Y} \pm t_{\alpha/2,n-1} S/\sqrt{n}$$

- Validating the model: (ε is a difference to allow valid decisions)
  - Suppose the C.I. does not contain μ_0:
    - If the best-case error is > ε, model needs to be refined.
    - If the worst-case error is ≤ ε, accept the model.
    - If best-case error is ≤ ε, additional replications are necessary.
  - Suppose the C.I. contains μ_0:
    - If either the best-case or worst-case error is > ε, additional replications are necessary.
    - If the worst-case error is ≤ ε, accept the model.

Confidence Interval Testing



Confidence Interval Testing

- Bank example: $\mu_0 = 4.3$, and "close enough" is $\varepsilon = 1$ minute of expected customer delay.
  - A *95%* confidence interval, based on the *6* replications is *[1.65, 3.37]* because:

$$\bar{Y} \pm t_{0.025,5} S / \sqrt{n}$$

$$2.51 \pm 2.571(0.82/\sqrt{6})$$

  - Falls outside the confidence interval,
    - the best case $|3.37 - 4.3| = 0.93 < 1$,
    - but the worst case $|1.65 - 4.3| = 2.65 > 1$,

➔ additional replications are needed to reach a decision.

Using Historical Output Data

- An alternative to generating input data:
  - Use the actual historical record.
  - Drive the simulation model with the historical record and then compare model output to system data.
  - In the bank example, use the recorded interarrival and service times for the customers $\{A_n, S_n, n = 1,2,\ldots\}$.
- Procedure and validation process: similar to the approach used for system generated input data.

Using a Turing Test

- Use in addition to statistical test, or when no statistical test is readily applicable.
- Utilize persons' knowledge about the system.
- For example:
  - Present 10 system performance reports to a manager of the system. Five of them are from the real system and the rest are "fake" reports based on simulation output data.
  - If the person identifies a substantial number of the fake reports, interview the person to get information for model improvement.
  - If the person cannot distinguish between fake and real reports with consistency, conclude that the test gives no evidence of model inadequacy.

Summary

- Model validation is essential:
  - Model verification
  - Calibration and validation
  - Conceptual validation
- Best to compare system data to model data, and make comparison using a wide variety of techniques.
- Some techniques that we covered (in increasing cost-to-value ratios):
  - Insure high face validity by consulting knowledgeable persons.
  - Conduct simple statistical tests on assumed distributional forms.
  - Conduct a Turing test.
  - Compare model output to system output by statistical tests.